Create Project in laraval

Definition:

Laravel is a PHP framework used to build modern web applications using clean, structured, and reusable code.

Instead of writing everything from scratch, Laravel provides a ready system to build web apps quickly

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Key Points:

- **Follows MVC (Model-View-Controller) architecture**
- **Makes development fast and easy**
- **Provides built-in features (routing, authentication, security)**
- **Uses Blade templating for UI**

Why Use Laravel?

- **Saves development time**
- **Organized code structure**
- **Secure and scalable applications**

Composer

Composer is a dependency manager in PHP used to install, manage, and update libraries or packages automatically.

Example:

composer require package-name

Installs a ready-made package in your project

In short Composer helps manage and install PHP libraries easily.

Key Points:

- **Installs external PHP packages**
- **Manages project dependencies**
- **Keeps libraries updated**
- **Required to install Laravel**

Why Use Composer?

- **Saves time**
- **Avoids writing code from scratch**
- **Easy package management**

Step 1: Requirements install karo

Sab se pehle ensure karo ke ye cheezein installed hon:

✅ **1. XAMPP**

- Apache + MySQL + PHP ke liye
- Download: https://www.apachefriends.org

✅ **2. Composer**

- Laravel install karne ke liye zaroori hai
- Check karne ke liye:

```
composer -V
```

✅ **3. PHP version**

Laravel 10/11 ke liye:

```
php -v
```

👉 PHP 8.1 ya us se upar hona chahiye

🚀 **Step 2: Laravel install karo**

Terminal (CMD) open karo aur htdocs folder me jao:

cd C:\xampp\htdocs

Ab Laravel project create karo:

composer create-project laravel/laravel school-management-system

👉 Ye command:

- Laravel download karegi
- Project folder bana degi

▶️ **Step 3: Project run karo**

Project folder me jao:

cd school-management-system

Server run karo:

php artisan serve

Browser me open karo:

http://127.0.0.1:8000

👉 Agar Laravel welcome page aa gaya = SUCCESS 🎉

⚙️ **Step 4: Database setup**

phpMyAdmin open karo:

http://localhost/phpmyadmin

New database banao:

school_db

.env file update karo:

Project folder me .env file open karo aur ye change karo:

```
DB_CONNECTION=mysql

DB_HOST=127.0.0.1

DB_PORT=3306

DB_DATABASE=clg_db

DB_USERNAME=root

DB_PASSWORD=
```

Root Folder Structure

```
school-management-system/
|
 ├── app/
 ├── bootstrap/
 ├── config/
 ├── database/
 ├── public/
 ├── resources/
 ├── routes/
 ├── storage/
 ├── tests/
 ├── vendor/
 ├── .env
```

```
├── artisan
├── composer.json
```

🔍 **Important folders (simple explanation)**

📁 **app/ 👉 (Main Logic Folder)**

👉 Yahan project ka main code hota hai

Iske andar:

- Models → database handle
- Controllers → logic

👉 Example:

Student.php (Model)
StudentController.php (Controller)

📁 **routes/ 👉 (Entry Point Logic)**

👉 Yahan decide hota hai:

- kaunsa URL → kaunsa controller

```
Route::get('/students', [StudentController::class, 'index']);
```

📁 **resources/ 👉 (UI / Frontend)**

👉 Yahan views hoti hain (HTML pages)

resources/views/

👉 Blade files:

index.blade.php

📁 **public/ 👉 (Entry Point Folder)**

👉 Ye browser se access hota hai

👉 Important file:

index.php

👉 Ye Laravel ka **main entry point** hai

📁 **database/ 👉 (Database related)**

- migrations → tables create
- seeders → test data

📁 **config/ 👉 (Settings)**

👉 Yahan project settings hoti hain:

- database
- mail
- cache

📁 **vendor/ 👉 (Composer packages)**

👉 Composer jo install karta hai sab yahan hota hai
👉 Isko kabhi manually edit nahi karte ❌

📁 **storage/ 👉 (Temporary data)**

👉 Logs, cache, uploads waghera

📁 **bootstrap/ 👉 (System start files)**

👉 Laravel ko start karne ke liye use hota hai

📄 **Important Files**

📄 **.env 👉 (Configuration file)**

👉 Yahan:

- database
- app settings

📄 **artisan 👉 (Command tool)**

👉 Laravel CLI tool

```
php artisan serve
php artisan migrate
```

📄 **composer.json 👉 (Dependencies list)**

👉 Kaun kaun se packages install hain

Deshboard Design for a Project

Step 1: Folder Structure (VERY IMPORTANT)

👉 Ye folders create karo:

```
resources/views/
|
 ├── layouts/
|   └── app.blade.php        👈 main layout
|
 ├── partials/
|    ├── navbar.blade.php
```

```
|   ├── sidebar.blade.php
|   └── footer.blade.php
|
├── students/
|   └── index.blade.php
```

1.resources/views/layouts/app.blade.php

```
<!DOCTYPE html>
<html>
<head>
   <title>@yield('title')</title>

   <style>
      body {
         margin:0;
         font-family: Arial;
         background:#f4f6f9;
      }

      /* NAVBAR */
      .navbar {
         background:#2c3e50;
         color:white;
         padding:15px 20px;
         display:flex;
         justify-content:space-between;
      }

      .navbar a {
         color:white;
         margin-right:15px;
         text-decoration:none;
      }
```

```
.navbar a:hover {
   text-decoration:underline;
}

/* LAYOUT */
.container {
   display:flex;
}

/* SIDEBAR */
.sidebar {
   width:220px;
   background:#1a252f;
   min-height:100vh;
   padding-top:20px;
}

.sidebar a {
   display:block;
   color:#ccc;
   padding:12px 20px;
   text-decoration:none;
}

.sidebar a:hover {
   background:#34495e;
   color:white;
}

/* CONTENT */
.content {
   flex:1;
}

/* PAGE TITLE */
.page-title {
   background:white;
```

```
            padding:15px;
            font-size:20px;
            font-weight:bold;
            border-bottom:1px solid #ddd;
        }

        /* MAIN CONTENT */
        .main-content {
            padding:20px;
        }

        .card {
            background:white;
            padding:20px;
            border-radius:8px;
            box-shadow:0 2px 5px rgba(0,0,0,0.1);
        }

        /* FOOTER */
        .footer {
            background:#2c3e50;
            color:white;
            text-align:center;
            padding:10px;
            margin-top:20px;
        }
    </style>
</head>

<body>

    @include('partials.navbar')

    <div class="container">

        @include('partials.sidebar')

        <div class="content">
```

```
        <div class="page-title">
            @yield('page-title')
        </div>

        <div class="main-content">
            <div class="card">
                @yield('content')
            </div>
        </div>

    </div>

  </div>

  @include('partials.footer')

</body>
</html>
```

2: Navbar

📄 resources/views/partials/navbar.blade.php

```
<div class="navbar">
  <div>
    <strong>🏫 School System</strong>
    <a href="/">Home</a>
    <a href="#">About</a>
    <a href="#">Contact</a>
  </div>

  <div>
    <a href="#">Logout</a>
```

```
    </div>
</div>
```

Sidebar

📄 resources/views/partials/sidebar.blade.php

```
<div class="sidebar">

    <a href="/students">👨‍🎓 Students</a>
    <a href="#">👩‍🏫 Teachers</a>
    <a href="#">📚 Courses</a>

</div>
```

Footer

📄 resources/views/partials/footer.blade.php

```
<div class="footer">
    <p>© 2026 School Management System</p>
</div>
```

Students Page (Better UI)

📄 resources/views/students/index.blade.php

```
@extends('layouts.app')

@section('title', 'Students')

@section('page-title', 'Students Management')

@section('content')

<h3>Students List</h3>
```

```
<table border="1" width="100%" cellpadding="10">
    <tr>
        <th>ID</th>
        <th>Name</th>
        <th>Email</th>
        <th>Class</th>
    </tr>

    <tr>
        <td>1</td>
        <td>Ali</td>
        <td>ali@gmail.com</td>
        <td>10th</td>
    </tr>

</table>

@endsection
```

Route add karo

routes/web.php

```
Route::get('/students', function () {
    return view('students.index');
});
```

Ab jab tum open karo:

http://127.0.0.1:8000/students

External php ini open cammad

C:\>\php-8.5.4\php.ini